

Background: Climate change has resulted in the increasing emergence of human infectious diseases (Baker et al. 2022). Over 70% of emerging infectious diseases are zoonotic (Jones et al. 2008), demonstrating that human health is intricately linked to environmental and wildlife health. Anthropogenic environmental changes such as pollution and habitat loss affect global climate patterns by increasing greenhouse gasses and reducing carbon sinks, leading to more extreme weather events and climbing global temperatures (Karl and Trenberth 2003). Alterations to temperature and moisture patterns can affect emerging infectious disease by contributing to more hospitable environments for sustaining pathogens or vectors (Epstein 2001; Vandegrift et al. 2010; Welch et al. 2019) and increasing the ability of pathogens and hosts to move to and persist in locations outside of the endemic range (Colón-González et al. 2021). These anthropogenic changes result in impacts over broad spatial scales. Moreover, small-scale human activities occurring at the human-wildlife interface are leading to increasing contact between humans and potential disease reservoirs, altering the risk of spillover of zoonotic infectious diseases (Gamble et al. 2023; Hassell et al. 2017). However, it is unknown how local-scale activities could be interacting with broad-scale anthropogenic effects like climate change to have a compounding negative impact on human health. In contrast to global climate change, which has multifaceted sources and stakeholders, local-scale activities can be more directly quantified for actionable changes that improve aspects of human health in response to emerging disease risks.

Backyard bird feeding and poultry keeping are two local-scale human activities that interact with global scale changes in climate to have considerable impacts to human health. Backyard bird feeding in the U.S. is a multibillion-dollar industry and nearly 70% of households engage in some form of wildlife feeding (US Department of the Interior 2022). However, human-supplemented food fundamentally alters the native avian community by decreasing avian species diversity, increasing density, and altering movement behavior (Robb et al. 2008). Human supplemented food also influences disease dynamics (Brown and Hall 2018) leading to increased exposure (e.g., mycoplasmal conjunctivitis in finches; Fischer et al. 1997) and altered infectious periods through changes to immune function and lowered mortality of infected birds (Owen et al. 2021). In addition to increased disease in birds, backyard bird feeding also provides a source of pathogens that spillover into people. Documented outbreaks of zoonotic disease in people have been tied to avian fecal material or dead birds at backyard feeders (CDC 2021). Although the immediate effects of food supplementation on birds and people may be local, we need to understand whether shifting environmental conditions may alter disease dynamics in ways that amplify these effects beyond local scales.

Secondly, the growing popularity of maintaining backyard poultry flocks in the United States (USDA 2013) could lead to increased frequency of spillover of avian infectious diseases between domestic and native bird populations (Ayala et al. 2020). Backyard flocks are low density groups of domesticated poultry owned by private citizens that are not regulated by the same biosecurity measures as poultry companies (USDA 2013). In particular, many backyard poultry are free range or accessible by wildlife, including vectors of zoonotic disease like mosquitoes. Poultry can thus harbor and transmit pathogens to native birds and people (Punchihewage-Don et al. 2022). Moreover, a survey by the USDA found that around 30% of backyard chicken owners also had bird feeders (USDA 2011), accentuating the risk of spillover events resulting from higher densities of wild birds at feeders and the potential for poultry to serve as reservoirs of infectious disease. However, no research has compared the

potential disease risk of these common backyard activities or investigated the potential mediating effects of aspects of climate change, despite the increasing number of families who have backyard flocks.

The aim of this proposed research is to assess disease dynamics of wild native bird communities in relation to local-scale human activities and climate change. We focus on *Salmonella enterica* and West Nile Virus (hereafter, “WNV”), two zoonotic infectious diseases that have a significant impact on human health and are also projected to be affected by climate change (Epstein 2001; Welch et al. 2019). Non-typhoidal *Salmonella* is the most common zoonotic food-borne pathogen in the U.S., and the leading bacterial cause of hospitalizations (Skarżyńska et al. 2021). Furthermore, antimicrobial resistant (AMR) genes in diverse bacterial pathogens, including *Salmonella*, are now a significant global burden costing millions of human lives each year (Murray et al. 2022). *Salmonella* can persist in a wide variety of host taxa, as well as in the environment. Evidence from multiple studies on environmental factors affecting *Salmonella* infections suggests that this bacterial pathogen is particularly problematic in warm seasons and regions (D’Souza et al. 2004; Welch et al. 2019). and selection favoring AMR genes is likely to intensify human health risks (Wang et al. 2025). Consequently, changes in contact frequency at the backyard human-wildlife interface driven by bird feeding and/or poultry keeping, as well as changes in temperature and moisture patterns due to climate change could alter the risk of spillover of *Salmonella* into humans.

Similarly, climate change is affecting the prevalence and geographic extent of mosquito-borne pathogens, like WNV (Colón-González et al. 2021). WNV was introduced to the US in 1999, where the impact of the outbreak was increased via climate change-driven mild winters followed by dry, hot summers (Epstein 2001). The primary vector of this virus is an urban-associated mosquito, *Culex pipiens*, such that anthropogenic change has led to WNV becoming the most severe vector-borne disease for humans in North America (Kilpatrick et al. 2006). Humans bitten by infected mosquitos can have adverse health effects, where a minority of cases result in severe neuroinvasive disease (Campbell et al. 2002). The rates of human infection are also projected to rise as temperatures continue to warm, extreme weather becomes more frequent or unpredictable and urbanization increases (Bradley et al. 2008; Harrigan et al. 2014). Additionally, several common bird species (northern cardinal, American robin, Carolina wren) have higher abundances in areas with bird feeders and are competent amplifying hosts for WNV (Kilpatrick et al. 2006, 2013). Local-scale human activities, coupled with broad-scale effects of climate change, could thus be increasing the risk of WNV spillover into people in their backyards.

To date, no research has evaluated the cumulative impact of bird feeders, backyard poultry, and temperature or moisture patterns on disease spillover risk from native wild birds. This is concerning because, as highly mobile organisms, wild birds connect large spatial scales and could be impactful agents transmitting disease and AMR genes among human and wildlife communities (De Lucia et al. 2018; Skarżyńska et al. 2021). Our goal with funding from the Burroughs Wellcome Fund Climate Change and Human Health Seed Grant is to analyze biological samples collected over multiple years and seasons to identify disease dynamics of these two common zoonotic infectious diseases at the backyard human-wildlife interface. We will use these empirical data to develop models that evaluate the conditions under which local-scale backyard activities and broad-scale effects of climate change increase zoonotic disease spillover risks to human health. **Therefore, this seed funding will have a**

large impact on informing the direction of future research on zoonotic disease risk at the human-wildlife interface with increasing effects of climate change.

Preliminary Data: We have biological (blood, fecal, soil) samples collected over 2 years from sites around eastern Alabama. We focused on backyards across four different treatments based on human activities: Households that have bird feeders, those that maintain small poultry flocks, those that have both bird feeders and poultry, and households that have neither. In addition, we sampled birds from natural areas that contain no human structures within 200m. This is an ongoing, year-round effort, such that we have samples across multiple seasons and years. We caught common resident avian species that are known hosts for WNV and *Salmonella* such as the northern cardinal, Carolina wren and house finch. However, we also caught migratory birds like neotropical warbler species, to gain information on the potential long-distance transmission of zoonotic disease. To date, we have caught over 250 birds, including a recapture rate of 8% across 15 sites. From each bird, we collected a blood sample stored on filter paper strips for later testing for WNV (see below). We also collected fecal and soil samples in an enrichment broth that were selectively cultured to evaluate presence of active *Salmonella* infections. Preliminary processing showed a 14% *Salmonella* positivity rate. Cultures from positive samples were preserved in enrichment broth in cryotubes at -80C until sequencing to determine serotype and presence of AMR genes (see below).

Hypothesis and Objectives: We hypothesize that local-scale human activities, such as maintaining backyard bird feeders and domesticated poultry, affect zoonotic disease dynamics of wild birds and that the effect is moderated by temperature and moisture patterns. We have three specific objectives: 1) We will use whole genome sequencing of *Salmonella* cultures from positive samples to determine whether it is a pathogenic serotype, whether AMR genes are present, and to quantify the extent to which these AMR genes are under selection in these *Salmonella* populations; 2) We will use ELISA and RT-qPCR to evaluate the avian blood samples for WNV infections; and 3) We will use agent-based models to test the influence of backyard activities and climate change on the prevalence of *Salmonella* and WNV in native birds to predict the risk of these diseases to human health.

Methods: We will leverage biological samples collected during our multi-year effort to evaluate zoonotic disease prevalence in native birds as a function of human activities and climate variables.

Salmonella enterica: Fecal samples from native birds and soil samples from each site were cultured using media selective for *Salmonella enterica*. Specifically, we enriched samples using buffered peptone water (BPW), then selectively enriched for *Salmonella* using a Rappaport-Vassiliadis broth. Subsequently, we streaked from the broth onto XLD agar to selectively grow *Salmonella* cultures if it was present in the sample. Cultures were then preserved in cryotubes containing BPW with glycerol and stored in a -80C freezer. Whole genome sequencing will be conducted on positive isolates using Illumina short-read sequencing. Genomic DNA will be extracted from archived isolates and prepared using standard library preparation protocols. We will target an average sequencing depth of ~50× coverage per isolate, which provides sufficient resolution for reliable genome assembly or mapping, serotype identification, and detection of AMR genes. Sequencing reads will be

demultiplexed and subjected to quality control, including adapter trimming and removal of low-quality (FastQC; Andrews 2010; Trimmomatic; Bolger et al. 2014). High-quality reads will then be mapped to a reference *Salmonella enterica* genome assembly (NCBI Refseq GCF_000006945.2) using a short-read aligner (BWA-MEM; Li and Durbin 2009). Resulting alignments will be sorted, duplicates removed, and indexed (SAMtools; Li et al. 2009), and then variants will be identified in the genome data (bcftools; Genovese et al. 2024), with filtering based on depth and quality thresholds to ensure high-confidence SNP identification. In parallel, genome assemblies will be generated using *de novo* assemblers (SPAdes; Bankevich et al. 2012) to support serotype confirmation and structural characterization. AMR genes will be identified by screening assembled genomes or mapped reads against curated databases (CARD; Alcock et al. 2023; ResFinder; Gschwind et al. 2023). In addition to presence/absence of AMR genes, we will conduct exploratory analyses to evaluate whether observed patterns are consistent with strong selection on resistance associated genes and loci. Specifically, we will consider patterns such as the recurrence of identical or highly similar AMR gene variants across independent isolates, reduced diversity in genomic regions linked to resistance loci, and potential clustering of AMR profiles by site type or environmental conditions. While sample sizes may limit formal inference of selection, these analyses will provide an initial indication of whether resistance traits are being non-randomly maintained or enriched in backyard-associated environments.

West Nile virus: During capture, we collected blood samples from native birds by pricking the ulnar wing vein with a sterile needle, then allowing blood to soak into Nobot filter paper strips. Once dry, these strips were stored at room temperature in the lab. To test for WNV exposure, a standardized portion of each dried blood spot will be excised and incubated in an elution buffer (phosphate-buffered saline) under controlled temperature and agitation conditions to rehydrate the sample and release antibodies into solution. Eluted samples will be screened for WNV using a commercial enzyme-linked immunosorbent assay (ELISA; ID Screen West Nile Competition Multi-Species, IDVet, Montpellier, France), which has previously been used to detect WNV antibodies in wild birds (Olofsson et al. 2024). For positive cases and where sample volume permits, RNA will be extracted using a commercially available kit designed for low-input samples (QIAmp Viral RNA Mini Kit, Qiagen), followed by RT-qPCR to test for active infection (Del Amo et al. 2013), again using protocols previously effective in wild birds (Sohier et al. 2026). Prevalence of WNV exposure will be summarized across backyard treatment categories, seasons, and years. We will then evaluate whether patterns vary as a function of local-scale human activities (bird feeders, poultry, both, or neither) and climate. Repeated sampling of sites will allow us to account for temporal variation and assess whether interannual differences in environmental conditions correspond to changes in WNV dynamics.

Climate variables: At each site (backyards and natural areas), we deployed a logger that continuously monitors temperature. We will pair these local specific data with past weather data sourced from NOAA Monthly U.S. Climate Divisional Database (NClimDiv) to estimate monthly average low and high temperature, humidity and precipitation. While these data occur at a broader scale than the temperature loggers at each site, our samples span the north-south extent of eastern Alabama and so these county-level climate data are relevant to the spatial scale of sampling.

Agent-based model: We developed BYPD (*Backyard Pathogen Dynamics*), an agent-based modeling framework designed to simulate interactions among multiple host species at backyard interfaces. The model includes common backyard bird species (Carolina chickadee (*Poecile carolinensis*), tufted titmouse (*Baeolophus bicolor*), Carolina wren (*Thryothorus ludovicianus*), northern cardinal (*Cardinalis cardinalis*), house finch (*Haemorhous mexicanus*), and backyard poultry). BYPD integrates observational and epidemiological data to elucidate the transmission dynamics of pathogens such as *Salmonella enterica* and WNV.

The model will be informed by empirical data inputs resulting from this proposed research on disease prevalence across environmental conditions, as well as additional data from our lab on contact networks and frequency of feeder use. Individual animals are represented as agents with species-specific behaviors, movement patterns, and social structure, allowing simulation of both direct and indirect transmission among wild birds and backyard poultry sharing the same environment. By incorporating heterogeneity in climate, human behavior, host abundance, management practices and pathogen characteristics, BYPD supports virtual experimentation and scenario testing. The framework can be used to explore the conditions that facilitate pathogen emergence and persistence, evaluate the effectiveness of alternative surveillance and intervention strategies, and inform real-world decision-making under realistic backyard conditions where multiple species coexist and interact (Belsare et al. 2020, 2022).

Conclusion: Outbreaks of zoonotic infectious diseases are projected to increase as a consequence of human population expansion, urbanization and climate change (Baker et al. 2022). To manage the accelerating frequency of outbreaks, we need additional research on zoonotic disease dynamics at the local scale in human-altered systems where outbreaks begin. Currently, no other research has quantified the impact on zoonotic disease prevalence of common activities at the backyard human-wildlife interface or the compounding influence of climate variability. The close association between humans and wild or domestic birds through maintaining bird feeders or poultry increases interspecific contact frequency and the potential for spillover and spillback of zoonotic disease (Lloyd-Smith et al. 2009). While spillover may initially have only a local impact, the practice of these backyard activities is extremely widespread such that they could be responsible for pervasive alterations to biodiversity and zoonotic disease dynamics.

The global-scale emergence of novel pathogens like COVID-19 lead to high morbidity and mortality; yet, frequent local outbreaks of endemic zoonotic diseases like *Salmonella* and WNV also result in a significant toll on human health and high economic costs (Kilpatrick et al. 2006; Murray et al. 2022). Moreover, by focusing on commonly reoccurring and harmful zoonotic pathogens within the timeline and funding amount of this grant, we can determine generalizable relationships between climate change, human activities and zoonotic disease. Through the identification of epidemiological and ecological aspects of these pathogens, and the use of an agent-based modeling framework, our results will inform responses to other vector-borne or bacterial zoonotic diseases with significant health impacts like Zika, dengue, *Staphylococcus aureus* or *Streptococcus pneumonia* (Bisanzio et al. 2021; Murray et al. 2022). Disease prevalence is affected by a multitude of local-scale and broad-scale human, biological and climate factors. Our study takes the first step in addressing the role of these interacting factors on two zoonotic infectious diseases that have large impacts to human health and are potentially present in people's backyards.

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